the design, development, tests and reviews. That's what more than sixty ISRO scientists did for five years alongside the French at the Vernon facility to develop the Vikas engine.

Some simpler systems, however, can be replicated once you have a close, long look, provided you know not just the design and theories, but also the material used. One such was the six-component static solid motor test facility we had access to in Istres. The concept of a solid motor test facility is simple: the motor or the rocket itself sans payload is kept in a horizontal position against a reinforced concrete wall. At the point of contact between the motor and the wall would be strain gauges. You fire the rocket motor which pushes against the concrete wall and the strain gauges pick up and measure the force exerted by the motor. Calculations give you the thrust to see if it is good enough to take the desired mass to the altitude of your choice.

At Thumba, we had a test facility which served our purpose of testing small solid boosters, but in comparison to the French, ours was at best primordial. A six-component test facility, the kind SEP had at its Istres centre could not just measure thrust, but also give accurate readings of variations when the rocket motor was tilted. In other words, it gave details of quantum and direction of the force. While allowed to roam around the Istres centre—we were in France, visiting several such facilities across two months—we were picking up these details.

Nandakumar, an electronics engineer from IIT who later set up the ISRO satellite tracking centres across India, took notes of measurements and instrumentation. I studied the location and functional details. On our return, we replicated the six-component facility at Sriharikota, which came to be known as STEX (Static Test and Evaluation Complex). This was later upgraded as the Vehicle Assembly, Static Test and Evaluation Complex (VAST).

But all these were just useful sideshows in my search for the right liquid engine. I kept pestering Morin. He showed me an engine called L-17, trying to sell it to ISRO. An outright technology transfer was not my idea; I wanted an ISRO group to work along French scientists to develop an